

Unit 3. Convexity, Bases, and Dimension

Convex sets

A subset S of a real vector space is *convex* if

$$p, q \in S, \quad 0 \leq t \leq 1 \quad \implies \quad (1-t)p + tq \in S.$$

Thus every segment with endpoints in S lies in S . The empty set and every one-point set are convex. A vector subspace is convex, but a convex set need not contain zero or be closed under arbitrary scalar multiplication.

Example 3.1 (Parallelograms and half-spaces). For fixed vectors u, v , the parallelogram

$$S = \{su + tv : 0 \leq s, t \leq 1\}$$

is convex. Indeed, the combination $(1-\lambda)(s_1u + t_1v) + \lambda(s_2u + t_2v)$ has coefficients $(1-\lambda)s_1 + \lambda s_2$ and $(1-\lambda)t_1 + \lambda t_2$, both in $[0, 1]$ when $0 \leq \lambda \leq 1$. For $a \in \mathbb{R}^n$ and $c \in \mathbb{R}$, the half-space $H = \{x : a \cdot x \geq c\}$ is convex: if $x, y \in H$, then

$$a \cdot ((1-t)x + ty) = (1-t)a \cdot x + ta \cdot y \geq c.$$

The same calculation works with $\leq$, or with either strict inequality. For example, each region $2x - 3y \geq 6$ and $2x - 3y \leq 6$ is convex.

A *convex combination* of $p_1, \dots, p_k$ is $\sum_{i=1}^k t_i p_i$ with $t_i \geq 0$ and $\sum_i t_i = 1$. Write

$$\text{conv}\{p_1, \dots, p_k\} = \left\{ \sum_{i=1}^k t_i p_i : t_i \geq 0, \sum_{i=1}^k t_i = 1 \right\}.$$

Theorem 3.1 (Convex hull). For a nonempty finite list of points, its set of convex combinations is convex and is contained in every convex set containing those points.

Proof. If $x = \sum_i a_i p_i$ and $y = \sum_i b_i p_i$ are convex combinations, then

$$(1-t)x + ty = \sum_i ((1-t)a_i + tb_i)p_i.$$

The new coefficients are nonnegative and have sum $(1-t) + t = 1$, proving convexity.

For the second assertion, let S be convex and contain every p_i . We prove by induction on k that every convex combination belongs to S . The case $k = 1$ is immediate. If $t_k = 1$, the combination equals p_k . Otherwise $1 - t_k > 0$, and

$$q = \sum_{i=1}^{k-1} \frac{t_i}{1-t_k} p_i$$

is a convex combination of $k-1$ points, so $q \in S$ by induction. Therefore $\sum_i t_i p_i = (1-t_k)q + t_k p_k \in S$. $\square$

For three noncollinear points, the convex hull is the filled triangle, including its edges. The span allows arbitrary real coefficients; an affine combination requires their sum to be one; a convex combination also requires nonnegative coefficients.

Linear independence, bases, and coordinates

A list $v_1, \dots, v_k$ is *linearly independent* if

$$c_1 v_1 + \dots + c_k v_k = 0 \implies c_1 = \dots = c_k = 0.$$

It is *dependent* if a relation exists with at least one nonzero coefficient. A relation is an equality of vectors, so for functions it must hold at every point of their domain. The empty list is independent. A list containing zero is dependent, since the coefficient of that vector can be one and all other coefficients zero.

Proposition 3.2 (Dependence and adjoining a vector). A finite list is dependent exactly when one of its vectors is a linear combination of the others. If $v_1, \dots, v_k$ is independent, then $v_1, \dots, v_k, w$ is independent exactly when $w \notin \text{span}(v_1, \dots, v_k)$.

Proof. In a nontrivial relation, choose $c_j \neq 0$ and solve $v_j = -\sum_{i \neq j} (c_i/c_j) v_i$. Conversely, moving such an expression to one side gives a relation whose coefficient of v_j is one. For the second assertion, a relation $\sum_i c_i v_i + dw = 0$ with $d = 0$ has all coefficients zero. Thus any nontrivial relation must have $d \neq 0$ and expresses w in the old span. $\square$

In particular, two vectors are dependent exactly when one is a scalar multiple of the other. The scalar may be zero; the criterion includes a pair consisting of zero and a nonzero vector.

A *basis* of V is an independent list that spans V . If $\mathcal{B} = (v_1, \dots, v_k)$ is an ordered basis and $v = \sum_i c_i v_i$, its *coordinate column* is $[v]_{\mathcal{B}} = (c_1, \dots, c_k)^{\top}$.

Proposition 3.3 (Unique coefficients). An independent list gives at most one expression for any vector as its linear combination. A basis gives exactly one expression for each vector in the space.

Proof. Subtract two expressions to obtain $\sum_i (c_i - d_i) v_i = 0$. Independence gives $c_i = d_i$ for every i . Spanning supplies existence for a basis. $\square$

Example 3.2 (A basis and its coordinate rule). Let $u = (1, 1)$ and $v = (-1, 2)$. Solving $\alpha u + \beta v = (s, t)$ gives

$$\alpha - \beta = s, \quad \alpha + 2\beta = t, \quad \alpha = \frac{2s + t}{3}, \quad \beta = \frac{t - s}{3}.$$

Every (s, t) has exactly one such expression, so (u, v) is a basis. In particular $[(1, 0)]_{(u, v)} = (2/3, -1/3)^{\top}$. Coordinates refer to the chosen basis and its order.

Proposition 3.4 (The two-coordinate test). The vectors (a, b) and (c, d) are independent if and only if $\Delta = ad - bc \neq 0$. When independent, they form a basis of $\mathbb{R}^2$.

Proof. If $\Delta \neq 0$, then the equations $a\alpha + c\beta = s$, $b\alpha + d\beta = t$ have the unique solution

$$\alpha = \frac{ds - ct}{\Delta}, \quad \beta = \frac{at - bs}{\Delta};$$

substitution verifies both equations. The zero right side proves independence, and arbitrary (s, t) proves spanning. If $\Delta = 0$ and $(a, c) \neq (0, 0)$, then the nonzero coefficient pair $(c, -a)$ gives the zero combination. If $a = c = 0$ and $(b, d) \neq (0, 0)$, use $(d, -b)$. If all four entries vanish, either vector alone gives a nontrivial relation. $\square$

Example 3.3 (Independence of functions). In the space of real functions on $\mathbb{R}$, suppose $ae^t + be^{2t} = 0$ for every t . Differentiating gives $ae^t + 2be^{2t} = 0$. Subtraction yields $be^{2t} = 0$, hence $b = 0$ and $a = 0$. Thus (e^t, e^{2t}) is a basis of its span, and the coordinates of $3e^t + 5e^{2t}$ are $(3, 5)^{\top}$.

Likewise $a \sin t + b \cos t = 0$ for every t gives $b = 0$ at $t = 0$ and $a = 0$ at $t = \pi/2$. These functions form a basis of their span. Testing one value would give only a necessary equation, not independence of the functions.

Example 3.4 (Matrices with equal row and column sums). Let W_n consist of the $n \times n$ real matrices whose row sums and column sums are all equal to the same number, which may vary with the matrix. The zero matrix belongs to W_n , and the common sum of $\alpha A + \beta B$ is $\alpha s_A + \beta s_B$; thus W_n is a subspace. For $n = 2$, equating row and column sums forces the form

$$\begin{pmatrix} a & b \\ b & a \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The two displayed matrices span W_2 and are independent by their first rows.

For $n = 3$, write the top-left four entries as a, b, c, d and the common sum as s . The first two row sums and column sums force

$$A = \begin{pmatrix} a & b & s - a - b \\ c & d & s - c - d \\ s - a - c & s - b - d & a + b + c + d - s \end{pmatrix}.$$

Both remaining sums now equal s . Put $e = s - a - b - c - d$ to obtain

$$A = \begin{pmatrix} a & b & c + d + e \\ c & d & a + b + e \\ b + d + e & a + c + e & -e \end{pmatrix} = aM_1 + bM_2 + cM_3 + dM_4 + eM_5,$$

where

$$M_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

$$M_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad M_5 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & -1 \end{pmatrix}.$$

Each M_i satisfies the required sums. These matrices span by the parameterization. A zero combination has entries $(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)$ equal to $a, b, c, d, -e$, forcing every coefficient to vanish. They therefore form a basis of W_3 .

Dimension

Theorem 3.5 (The generator bound). If V is spanned by m vectors, then every list of more than m vectors in V is dependent.

Proof. Let $v_1, \dots, v_m$ span V and let $w_1, \dots, w_k \in V$ with $k > m$. Choose coefficients a_{ij} such that $w_j = \sum_{i=1}^m a_{ij}v_i$. The homogeneous system

$$\sum_{j=1}^k a_{ij}c_j = 0 \quad (i = 1, \dots, m)$$

has a nonzero solution, because it has more unknowns than equations. For that choice,

$$\sum_{j=1}^k c_j w_j = \sum_{i=1}^m \left(\sum_{j=1}^k a_{ij}c_j \right) v_i = 0.$$

Thus the w_j are dependent. The generators v_i need not be independent: vanishing of the inner coefficients is sufficient for the relation. If $m = 0$, then $V = \{0\}$ and the assertion follows directly. $\square$

Theorem 3.6 (Existence and size of bases). Every space with a finite spanning list has a basis obtained from that list. All its bases have the same size.

Proof. Process a finite spanning list in order. Keep a vector exactly when it is outside the span of the previously kept vectors. Proposition 3.2 shows that the kept list is independent. Each discarded vector is in its span, so the kept list still spans V . For $V = \{0\}$ the kept list is empty.

If two bases have sizes m and n , the first spans and the second is independent, so Theorem 3.5 gives $n \leq m$. Interchanging their roles gives $m \leq n$, hence equality. $\square$

A space with a finite basis is *finite dimensional*; its *dimension* is the number of basis vectors. In particular, $\dim\{0\} = 0$, and $\dim \mathbb{R}^n = n$ because the standard vectors $e_1, \dots, e_n$ form a basis.

Theorem 3.7 (Extension and subspaces). Let $\dim V = n$.

- (a) Every independent list in V extends to a basis of V .
- (b) An independent list of n vectors is a basis. A spanning list of n vectors is a basis.
- (c) Every subspace $W \subseteq V$ has a finite basis and $\dim W \leq n$. Equality holds exactly when $W = V$.

Proof. (a) Append a fixed basis of V to the given independent list and apply the deletion procedure in Theorem 3.6. No original vector is discarded, and the resulting list is a basis.

(b) An independent list of size n cannot acquire another vector under extension, because all bases have size n . A spanning list can be reduced to a basis, but none of its n vectors can be discarded without producing a basis of size less than n .

(c) Starting with the empty list, choose a vector in W outside the span of those already chosen whenever possible. Every choice preserves independence. The generator bound prohibits more than n choices, so the procedure stops and its list spans W . Its size is at most n . If its size is n , part (b) makes this list a basis of V , giving $V \subseteq W$. The converse is immediate. $\square$

Example 3.5 (Removing a redundant vector and extending a basis). Consider the columns a_1, a_2, a_3 of

$$A = \begin{pmatrix} 1 & 2 & 3 \\ -2 & 3 & 8 \\ 5 & 1 & -3 \\ -3 & -4 & -5 \end{pmatrix}, \quad W = \text{span}(a_1, a_2, a_3).$$

The relation $a_1 - 2a_2 + a_3 = 0$ removes a_3 . If $sa_1 + ta_2 = 0$, the first two coordinates give $s + 2t = 0$, $-2s + 3t = 0$, hence $s = t = 0$. Therefore (a_1, a_2) is a basis of W and $\dim W = 2$.

Extend it by e_3, e_4 . In $sa_1 + ta_2 + ue_3 + ve_4 = 0$, the first two coordinates still give $s = t = 0$; the last two then give $u = v = 0$. These four vectors are independent in $\mathbb{R}^4$, hence form a basis. More explicitly, for $x = (x_1, x_2, x_3, x_4)^\top$,

$$s = \frac{3x_1 - 2x_2}{7}, \quad t = \frac{2x_1 + x_2}{7}, \quad u = x_3 - 5s - t, \quad v = x_4 + 3s + 4t$$

give its coordinates in this extended basis. The coefficients s, t are not themselves coordinates for an arbitrary vector relative to (a_1, a_2) unless $u = v = 0$.

Example 3.6 (Dimensions of matrix and polynomial spaces). Let E_{ij} have a one in position (i, j) and zeros elsewhere. Every $m \times n$ matrix has the unique expression $A = \sum_{i,j} a_{ij} E_{ij}$, so $\dim \mathbb{R}^{m \times n} = mn$. For diagonal matrices, use $E_{11}, \dots, E_{nn}$. For upper triangular matrices, use E_{ij} with $i \leq j$, giving dimension $n(n+1)/2$.

A symmetric matrix has the unique expression

$$A = \sum_i a_{ii} E_{ii} + \sum_{i < j} a_{ij} (E_{ij} + E_{ji}).$$

Thus the displayed matrices form a basis and the symmetric subspace has dimension $n + n(n-1)/2 = n(n+1)/2$.

The polynomials of degree at most d , together with the zero polynomial, have basis $1, t, \dots, t^d$. Spanning follows from the polynomial expression. If $\sum_{j=0}^d c_j t^j = 0$ for every real t , evaluating its j th derivative at zero gives $j!c_j = 0$. Thus the basis is independent and this space has dimension $d+1$. The space of all polynomials is infinite dimensional, since it contains independent lists of arbitrarily large size.

Exercises and solutions

Suggested practice: 3.6, 3.8, 3.10, 3.13, 3.15, 3.17, 3.20, 3.21, 3.24.

Exercise 3.1 (Parallelograms). Prove that $\{su + tv : 0 \leq s, t \leq 1\}$ is convex for any two fixed vectors u, v .

Solution. For points $x = s_1u + t_1v$ and $y = s_2u + t_2v$ and $\lambda \in [0, 1]$, their combination has coefficients $(1-\lambda)s_1 + \lambda s_2$ and $(1-\lambda)t_1 + \lambda t_2$. Each is nonnegative and at most $(1-\lambda) + \lambda = 1$, so the combination belongs to the set. No independence of u, v is required.

Exercise 3.2 (Half-spaces). Let $a \neq 0$ in $\mathbb{R}^n$ and $c \in \mathbb{R}$. Prove that $\{x : a \cdot x \geq c\}$ is convex.

Solution. If $a \cdot x, a \cdot y \geq c$ and $t \in [0, 1]$, then $a \cdot ((1-t)x + ty) = (1-t)a \cdot x + ta \cdot y \geq (1-t)c + tc = c$. This is the defining convexity condition.

Exercise 3.3 (Scaling a convex set). If S is convex and $c \in \mathbb{R}$, prove that $cS = \{cs : s \in S\}$ is convex.

Solution. For $x = cs_1, y = cs_2 \in cS$, we have $(1-t)x + ty = c((1-t)s_1 + ts_2) \in cS$. This proof includes negative c and $c = 0$, and does not divide by c . If S is empty, so is cS , and convexity is vacuous.

Exercise 3.4 (Intersection). Prove that the intersection of two convex sets is convex.

Solution. If $x, y \in S \cap T$, their segment lies in S and in T , hence in $S \cap T$. The same argument is valid when the intersection is empty: there are no endpoints for which the condition could fail.

Exercise 3.5 (Translation). If S is convex and w is fixed, prove that $w + S$ is convex.

Solution. For $x = w + s_1, y = w + s_2$, the combination $(1-t)x + ty = w + ((1-t)s_1 + ts_2)$ belongs to $w + S$, because the coefficients of w sum to one.

Exercise 3.6 (Dependence criteria). Prove the following for finite lists.

- (a) A list consisting only of zero is dependent; any list containing zero is dependent.
- (b) A list of nonzero vectors is dependent if and only if one vector is a linear combination of the others.
- (c) A list of nonzero vectors $x_1, \dots, x_n$ is dependent if and only if some x_j , $2 \leq j \leq n$, is a linear combination of $x_1, \dots, x_{j-1}$.

Solution. (a) Give the zero vector coefficient one and all other vectors coefficient zero. (b) In a nontrivial relation choose a nonzero coefficient and solve for its vector. Conversely, an expression for one vector gives a relation with coefficient one on that vector. (c) Given a nontrivial relation, let j be its largest index with nonzero coefficient. The hypothesis $x_1 \neq 0$ excludes $j = 1$. Solving for x_j expresses it in terms of its predecessors. Conversely, move such an expression to one side and assign coefficient zero to every later vector.

Exercise 3.7 (Column and row relations). Prove that the columns of I_n are independent. For

$$A = \begin{pmatrix} 1 & 3 & 2 & 0 \\ 1 & -5 & -3 & 1 \\ 5 & -1 & 0 & 2 \end{pmatrix},$$

show that its columns are dependent and determine whether its rows are dependent.

Solution. The equation $\sum_i c_i e_i = 0$ has coordinate i equal to c_i , so all coefficients vanish. If a_j denotes column j of A , then $3a_1 - a_2 - 8a_4 = 0$, a nontrivial relation. If r_i denotes row i , then $3r_1 + 2r_2 - r_3 = 0$. Thus both lists are dependent. These two calculations concern different lists of vectors, in $\mathbb{R}^3$ and $\mathbb{R}^4$ respectively.

Exercise 3.8 (Triangular independence). Prove that the columns of a square triangular matrix are independent if and only if every diagonal entry is nonzero.

Solution. For a lower triangular matrix A , if every diagonal entry is nonzero, the equations $Ax = 0$ successively force $x_1 = 0, x_2 = 0, \dots, x_n = 0$. For the converse, choose the largest index i with $a_{ii} = 0$. Set $x_1 = \dots = x_{i-1} = 0$ and $x_i = 1$. The first i equations hold. Each subsequent equation has a nonzero diagonal coefficient, so it determines the next x_j . The resulting vector is nonzero and satisfies $Ax = 0$. For an upper triangular matrix, run the first argument from the last row upward. If a diagonal entry is zero, choose the smallest such index i , set $x_{i+1} = \dots = x_n = 0, x_i = 1$, and solve the earlier equations upward; their diagonal entries are nonzero.

Exercise 3.9 (Pairwise sums). If x, y, z are independent real vectors, are $x+y, x+z, y+z$ independent?

Solution. Yes. A zero combination has coefficients $\alpha + \beta, \alpha + \gamma, \beta + \gamma$ on x, y, z , respectively. Independence makes these three sums zero. The first two give $\beta = \gamma = -\alpha$, and the third gives $-2\alpha = 0$. Thus all three coefficients vanish.

Exercise 3.10 (A parameter-dependent list). For which real λ are $(\lambda, 1, 0)^\top, (1, \lambda, 1)^\top$, and $(0, 1, \lambda)^\top$ dependent? Exhibit a nontrivial relation in each exceptional case.

Solution. Coefficients α, β, γ must satisfy $\lambda\alpha + \beta = 0, \alpha + \lambda\beta + \gamma = 0, \beta + \lambda\gamma = 0$. The first two give $\beta = -\lambda\alpha$ and $\gamma = (\lambda^2 - 1)\alpha$. The last becomes $\lambda(\lambda^2 - 2)\alpha = 0$. If $\lambda \notin \{0, \sqrt{2}, -\sqrt{2}\}$, then $\alpha = \beta = \gamma = 0$. For $\lambda = 0$, take $(\alpha, \beta, \gamma) = (1, 0, -1)$. For $\lambda = \pm\sqrt{2}$, take $(1, -\lambda, 1)$. These relations prove all the stated exceptions.

Exercise 3.11 (A finite span of random variables). Let $X_1, \dots, X_n$ be random variables with finite second moments, identifying random variables that are equal with probability one. Show that the vector space $V = \{\sum_{i=1}^n a_i X_i : a_i \in \mathbb{R}\}$ is finite dimensional.

Solution. The listed random variables span V by its definition. The basis-extraction theorem gives a basis selected from this list, so $\dim V \leq n$. Equality need not hold, because the X_i may be dependent as vectors. Finite second moments ensure the stated class is closed: pointwise Cauchy–Schwarz gives $(\sum_i a_i X_i)^2 \leq (\sum_i a_i^2)(\sum_i X_i^2)$, whose expectation is finite. No covariance-rank assertion is needed.

Exercise 3.12 (Infinitely many bases). Find two different bases of $\mathbb{R}^3$ and prove that $\mathbb{R}^3$ has infinitely many bases.

Solution. The standard basis is one. For every $\alpha \neq 0$, the three columns

$$b_1 = (1, 0, 2)^\top, \quad b_2 = (0, 2, 0)^\top, \quad b_3 = (0, 0, \alpha)^\top$$

form a basis: every x has the unique expression $x = x_1 b_1 + (x_2/2) b_2 + ((x_3 - 2x_1)/\alpha) b_3$. Taking $\alpha = 1$ supplies a second basis, and varying α over positive reals supplies infinitely many distinct bases.

Exercise 3.13 (Selecting a basis). Let $a_1, \dots, a_9$ be the columns of

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 5 \\ 0 & 1 & 1 & 2 & 2 & 3 & -1 & 3 & 3 \\ 1 & 1 & 2 & 3 & 5 & 5 & 1 & 0 & 4 \end{pmatrix}.$$

Which lists form a basis of $\mathbb{R}^3$?

- (a) a_1, a_2 ;
- (b) a_1, a_4, a_6, a_8 ;
- (c) a_2, a_4, a_7 ;
- (d) a_3, a_5, a_9 .

Solution. (a) No: two vectors cannot span a space of dimension three. (b) No: four vectors in $\mathbb{R}^3$ are dependent. (c) Yes. A zero combination with coefficients x, y, z gives $x + y + 2z = 0$, $x + 2y - z = 0$, $x + 3y + z = 0$. Subtracting the first equation from the second gives $y - 3z = 0$; subtracting the second from the third gives $y + 2z = 0$. Hence $z = y = x = 0$. Three independent vectors form a basis. (d) No: $7a_3 - 2a_5 - a_9 = 0$ is a nontrivial relation.

Exercise 3.14 (Constructing bases). In a finite-dimensional vector space, prove that any spanning set contains a basis and that any independent list can be extended to a basis. Deduce existence of a basis in the nonzero finite-dimensional case from a maximal independent list.

Solution. First choose a finite spanning list of the ambient finite-dimensional space. If S is any spanning set, each vector of that list is a finite combination of elements of S . Their combined finitely many elements form a finite spanning list drawn from S . Scan this list in order, discarding every vector in the span of those kept. The kept list is independent by the adjoining criterion. Each discarded vector is in its span, so spanning is preserved. For extension, place the given independent list first and append a fixed finite spanning list. Apply the same procedure: no vector of the initial list is discarded. Finally, a maximal independent list spans, because any vector outside its span could be adjoined while preserving independence. A bound on the size of independent lists ensures that a maximal list can be obtained after finitely many adjunctions.

Exercise 3.15 (Comparing sizes). Suppose $x_1, \dots, x_n$ spans V and $y_1, \dots, y_m$ is independent in V .

- (a) Prove $m \leq n$ without assuming the x_i are independent.
- (b) Deduce that every basis of a finite-dimensional space has the same size.
- (c) If $\dim V = d$, show that more than d vectors cannot be independent and fewer than d vectors cannot span.

Solution. (a) Write $y_j = \sum_i a_{ij}x_i$. If $m > n$, the homogeneous system $\sum_j a_{ij}c_j = 0$ has a nonzero solution c . It gives $\sum_j c_j y_j = 0$, a contradiction. (b) Use (a) twice, interchanging which basis is the spanning list and which is independent. (c) Compare an independent list to a basis of size d for the first claim, and compare a spanning list to that basis as an independent list for the second.

Exercise 3.16 (Coordinates and full dimension). Let $\dim V = n$.

- (a) Show that each vector has unique coordinates relative to a given basis.
- (b) Show that every independent list of n vectors is a basis, and that every basis has n vectors.
- (c) For a subspace $W \subseteq V$, prove $\dim W \leq n$, with equality if and only if $W = V$.

Solution. (a) Spanning supplies coordinates; subtracting two expressions and using independence proves uniqueness, including for zero. (b) If an independent list of n vectors failed to span, a vector outside its span could be adjoined, contradicting the generator bound. All bases have size n by the equal-size theorem. (c) Select vectors of W outside the span of previous choices. The process stops after at most n choices and gives a basis of W . If there are n choices, part (b) makes it a basis of V , so $V = W$. If $V = W$, their dimensions are equal by definition.

Exercise 3.17 (A sum of subspaces). Let U, W be finite-dimensional subspaces. Suppose finite lists A and B span U and W , respectively.

- (a) Prove that their combined list spans $U + W$.
- (b) Assuming $U \cap W = \{0\}$, prove $\dim(U + W) = \dim U + \dim W$ by first choosing bases from the two spanning lists.

Solution. (a) Every $u + w$ is the sum of a combination from A and a combination from B , hence a combination of the combined list. Conversely such combinations belong to $U + W$. (b) Extract bases $u_1, \dots, u_r$ and $w_1, \dots, w_s$. In a relation $\sum_i a_i u_i + \sum_j b_j w_j = 0$, the vector $\sum_i a_i u_i = -\sum_j b_j w_j$ lies in $U \cap W$, hence equals zero. Independence of each basis gives all $a_i, b_j = 0$. The combined list is therefore a basis of $U + W$ with $r + s$ elements. The original spanning lists need not themselves be bases.

Exercise 3.18 (Numerical independence). Prove independence of each list:

- (a) $(1, 1, 1), (0, 1, -2)$;
- (b) $(1, 0), (1, 1)$;
- (c) $(-1, 1, 0), (0, 1, 2)$;
- (d) $(2, -1), (1, 0)$;
- (e) $(\pi, 0), (0, 1)$;
- (f) $(1, 2), (1, 3)$;
- (g) $(1, 1, 0), (1, 1, 1), (0, 1, -1)$;
- (h) $(0, 1, 1), (0, 2, 1), (1, 5, 3)$.

Solution. In (a), the first and second coordinates of a zero combination force its two coefficients to be zero. In (b), use the second and then first coordinates. In (c), use the first and third. In (d), use the second and first. In (e), use both coordinates and $\pi \neq 0$. In (f), $\alpha + \beta = 0$ and $2\alpha + 3\beta = 0$ imply $\beta = \alpha = 0$. In (g), coefficients α, β, γ give $\alpha + \beta = 0$, $\alpha + \beta + \gamma = 0$, $\beta - \gamma = 0$. Hence $\gamma = \beta = \alpha = 0$. In (h), the first coordinate gives $\gamma = 0$; the others give $\alpha + 2\beta = 0$ and $\alpha + \beta = 0$, so $\beta = \alpha = 0$.

Exercise 3.19 (Two-dimensional coordinates). Find the coordinates of x relative to the ordered pair (a, b) in each case:

- (a) $x = (1, 0)$, $a = (1, 1)$, $b = (0, 1)$;
- (b) $x = (2, 1)$, $a = (1, -1)$, $b = (1, 1)$;
- (c) $x = (1, 1)$, $a = (2, 1)$, $b = (-1, 0)$;
- (d) $x = (4, 3)$, with the same a, b as in (c).

Solution. Solve $\alpha a + \beta b = x$. In (a), $\alpha = 1$, $\alpha + \beta = 0$, giving $(1, -1)^\top$. In (b), $\alpha + \beta = 2$, $-\alpha + \beta = 1$, giving $(1/2, 3/2)^\top$. In (c), $\alpha = 1$ and $2\alpha - \beta = 1$, giving $(1, 1)^\top$. In (d), $\alpha = 3$ and $2\alpha - \beta = 4$, giving $(3, 2)^\top$. In each case the coefficient equations with zero right side have only the zero solution, so the ordered pair is a basis and these coordinates are unique.

Exercise 3.20 (Three-dimensional coordinates). Find coordinates in the indicated ordered basis:

- (a) $x = (1, 0, 0)$; $a = (1, 1, 1)$, $b = (-1, 1, 0)$, $c = (1, 0, -1)$;
- (b) $x = (1, 1, 1)$; $a = (0, 1, -1)$, $b = (1, 1, 0)$, $c = (1, 0, 2)$;
- (c) $x = (0, 0, 1)$, with the basis in (a).

Solution. For the basis in (a), the equations are $\alpha - \beta + \gamma = x_1$, $\alpha + \beta = x_2$, $\alpha - \gamma = x_3$. Thus $3\alpha = x_1 + x_2 + x_3$, $\beta = x_2 - \alpha$, $\gamma = \alpha - x_3$. Parts (a) and (c) give $(1/3, -1/3, 1/3)^\top$ and $(1/3, -1/3, -2/3)^\top$. For (b), $\beta + \gamma = 1$, $\alpha + \beta = 1$, $-\alpha + 2\gamma = 1$ give $\gamma = 1$, $\alpha = 1$, $\beta = 0$. The coordinates are $(1, 0, 1)^\top$. Setting the right sides to zero in either system forces all three coefficients to be zero, verifying independence of each basis.

Exercise 3.21 (Independence of functions on $\mathbb{R}$). Prove that each pair is independent as real functions on $\mathbb{R}$:

- (a) $1, t$;
- (b) t, t^2 ;
- (c) t, t^4 ;
- (d) e^t, t ;
- (e) te^t, e^{2t} ;
- (f) $\sin t, \cos t$;
- (g) $t, \sin t$;
- (h) $\sin t, \sin 2t$;
- (i) $\cos t, \cos 3t$.

Solution. Write $af(t) + bg(t) = 0$ for every t . (a) At 0 and 1, obtain $a = 0$ and then $b = 0$. (b) At 1 and -1 , obtain $a + b = 0$ and $-a + b = 0$. (c) The same two values give $a + b = 0$ and $-a + b = 0$. (d) At 0, obtain $a = 0$; at 1, obtain $b = 0$. (e) At 0, obtain $b = 0$; at 1, obtain $a = 0$. (f) Use 0 and $\pi/2$. (g) At π , obtain $a\pi = 0$; at $\pi/2$, obtain $b = 0$. (h) At $\pi/2$, obtain $a = 0$; at $\pi/4$, obtain $b = 0$. (i) At 0 and $\pi/3$, obtain $a + b = 0$ and $a/2 - b = 0$, giving $a = b = 0$.

Exercise 3.22 (Functions on the positive half-line). Prove independence on $t > 0$ of (a) $t, 1/t$ and (b) $e^t, \log t$.

Solution. (a) A relation $at + b/t = 0$ gives $a + b = 0$ at $t = 1$ and $2a + b/2 = 0$ at $t = 2$. Multiplying the second equation by two and subtracting the first yields $3a = 0$, hence $b = 0$. (b) At $t = 1$, a relation $ae^t + b \log t = 0$ gives $ae = 0$, so $a = 0$. At $t = e$, it then gives $b = 0$. Both test values belong to the specified domain.

Exercise 3.23 (Coordinates of a derivative). Find the coordinates of $f(t) = 3 \sin t + 5 \cos t$ and its derivative relative to the ordered basis $(\sin t, \cos t)$ of their span.

Solution. The coordinate columns are $[f] = (3, 5)^\top$ and $[f'] = (-5, 3)^\top$, because $f'(t) = -5 \sin t + 3 \cos t$. Independence of sine and cosine proves uniqueness of both expressions.

Exercise 3.24 (Bases of matrix spaces). Exhibit bases, prove their basis properties, and compute dimensions for:

- (a) all 2×2 real matrices and, more generally, all $m \times n$ real matrices;
- (b) diagonal $n \times n$ matrices;
- (c) upper triangular $n \times n$ matrices;
- (d) symmetric 2×2 , 3×3 , and general $n \times n$ matrices.

Solution. Write E_{ij} for the matrix unit. In (a), the basis is all E_{ij} , since $A = \sum_{i,j} a_{ij} E_{ij}$ and the coefficient of each unit is fixed by the corresponding entry. The dimensions are 4 and mn . In (b), use $E_{11}, \dots, E_{nn}$, giving dimension n . In (c), use the E_{ij} with $i \leq j$; the number is $n + (n-1) + \dots + 1 = n(n+1)/2$. In (d), use E_{ii} for $1 \leq i \leq n$ and $E_{ij} + E_{ji}$ for $i < j$. A symmetric matrix is their combination with coefficients a_{ii}, a_{ij} . If the combination is zero, its diagonal and upper triangular entries force all coefficients zero. There are $n + n(n-1)/2$ basis matrices, giving dimensions 3, 6, and $n(n+1)/2$, respectively.

Exercise 3.25 (Classifying low-dimensional subspaces). Classify the subspaces of $\mathbb{R}^2$ and of $\mathbb{R}^3$ using dimension.

Solution. A subspace of $\mathbb{R}^2$ has dimension zero, one, or two. Dimension zero means $\{0\}$; dimension one means a line $\{tu : t \in \mathbb{R}\}$ with $u \neq 0$; dimension two means the whole space by the full-dimension theorem. A subspace of $\mathbb{R}^3$ has dimension zero, one, two, or three. The first two cases are zero and a line through zero. Dimension two gives $\{su + tv : s, t \in \mathbb{R}\}$ for independent u, v , a plane through zero. Dimension three gives all of $\mathbb{R}^3$. Each described set is indeed a subspace, since it is a span.

Exercise 3.26 (Arbitrarily long independent lists). In the real vector space of polynomials of all degrees, prove that $1, t, \dots, t^n$ are independent for every positive integer n . Deduce that the space is infinite dimensional.

Solution. If $\sum_{j=0}^n a_j t^j = 0$ for every real t , its k th derivative at zero is $k!a_k = 0$, for $0 \leq k \leq n$. Hence all coefficients vanish. If the space had finite dimension d , it could not contain the independent list $1, t, \dots, t^d$ of size $d+1$. Thus no finite basis exists.